Full Apartment Composition & Budget Logic

AID-LOG-007 | Revision 1

Purpose. Defines the project-level logic used by AIDOOi to coordinate room modules, preserve room-specific fit requirements, allocate a total apartment budget, maintain cross-room design coherence, and generate a unified whole-apartment recommendation.

> **Core principle: Full Apartment Logic coordinates room-level recommendations; it does not replace or duplicate room-level LOG standards.**

1. Project Architecture

A Full Apartment is a coordinated multi-room project. Each included room is first resolved through its own room-level logic, then AID-LOG-007 applies a project coordination layer.

Room Module	Referenced Logic	Project Role
Bedroom(s)	AID-LOG-003	Repeated for the number of bedrooms entered in AID-SPC-007.
Living Room	AID-LOG-002	Required apartment module.
Dining Room	AID-LOG-004	Required apartment module.
Home Office	AID-LOG-005	Included only when selected by the customer.

Total Apartment Area is not required for the MVP. Project scale is driven primarily by the number of bedrooms and included room modules, while each room's own dimensions determine its composition and fit.

2. Cross-Room Coordination Logic

Overall Style is the Project Design Anchor. All rooms should remain within the selected style direction while retaining their own functional character.

Overall Colour defines a coordinated project palette, not a requirement for every product to use the same colour.

> **Cross-room controls: Style Coherence + Colour Harmony + Material Compatibility + Overall Budget Balance.**

Rooms may express different functional character while remaining visually coherent within the overall project direction. The objective is consistency without duplication.

3. Bedroom Consistency Preference

When an apartment contains two or more bedrooms, AID-SPC-007 asks how the customer wants the bedrooms furnished.

Preference	Logic
Same Products	Use the same furniture products across bedrooms where room dimensions, bed size, fit and availability allow.
Different Products	Use different products while maintaining the same overall Style, Colour Direction and project design coherence.

Rule: Same Products is a preference, not a fit override. AIDOOi must not compromise bed size, circulation or room fit merely to repeat an identical product.

Same Products does not imply an assumed quantity discount. A discount is applied only when it exists in the actual seller offer.

4. Project Budget Allocation Logic

Sequence: Total Project Budget -> Included Room Modules -> Room Composition Requirements -> Relative Cost Weight -> Dynamic Budget Allocation.

The budget is not divided using fixed room percentages. Allocation is determined dynamically from the actual composition and requirements of each room.

Project Budget has priority over forcing the same Budget Tier in every room. A project may mix product and room-level price tiers to optimize the complete apartment.

Allocation protection order: Preserve Fit -> Preserve Essential Furniture -> Adjust Product Tiers -> Reduce Conditional / Secondary Items -> Reduce Decor.

If the project still cannot satisfy essential room requirements within the selected total budget, AIDOOi should surface a Budget Constraint rather than silently remove essential furniture.

5. Full Apartment Project Budget Ranges

These ranges are total furnishing targets for the base apartment project. Home Office is excluded from the base range and is handled as a modifier. For 5+ bedroom apartments, the 4BR base is extended using Additional Bedroom modules.

Apartment Scale	Essential	Value	Standard	Premium	Luxury
1 Bedroom	AED 7,000-12,000	AED 12,000-18,000	AED 18,000-28,000	AED 28,000-45,000	AED 45,000+
2 Bedrooms	AED 10,000-17,000	AED 17,000-26,000	AED 26,000-40,000	AED 40,000-65,000	AED 65,000+
3 Bedrooms	AED 13,000-22,000	AED 22,000-34,000	AED 34,000-52,000	AED 52,000-82,000	AED 82,000+
4 Bedrooms - Base	AED 16,000-27,000	AED 27,000-42,000	AED 42,000-64,000	AED 64,000-100,000	AED 100,000+

Apartment Budget Range Rule: The range is the initial project-level target. Final allocation is dynamically adjusted according to actual room dimensions, bed sizes, compositions and included modules.

Budget Tier Rule: A Project Budget Tier defines a target range, not a requirement to spend the full amount. Do not add unnecessary products or upgrades merely to reach a lower boundary.

6. Project Budget Modifiers

Modifier	Application
Additional Bedroom	For apartments above 4 bedrooms, each additional bedroom is calculated from its actual AID-LOG-003 composition and added to the 4BR base project budget.
Home Office	If Home Office = Yes, its cost is calculated using AID-LOG-005 according to room size, workspace users and composition, then added to the project budget.

Bedroom Consistency Preference is a Recommendation Modifier, not a Budget Modifier.

7. Final Recommendation Logic

Apartment Scale by Bedroom Count -> Room Modules -> Room Fit & Composition -> Bedroom Consistency Preference -> Project Budget + Modifiers -> Cross-Room Style / Colour Coordination -> Product Selection -> Whole-Apartment Recommendation.

AIDOOi optimizes the apartment as one coordinated project while preserving the functional and fit requirements of each individual room.

Room-level Product Price Tiers may be mixed when doing so creates a better complete-project result within the selected total budget.

Final Project Price is the sum of Current Selling Prices of all selected products and should recalculate dynamically after removals or substitutions.

8. Standard References

Reference	Standard
AID-SPC-007	Full Apartment Path Standard
AID-LOG-001	Product Price Range Standard
AID-LOG-002	Living Room Composition, Fit & Budget Logic
AID-LOG-003	Bedroom Composition, Fit & Budget Logic
AID-LOG-004	Dining Room Composition, Fit & Budget Logic
AID-LOG-005	Home Office Composition, Fit & Budget Logic
AID-DATA-007	Product Colour Standard
AID-DATA-009	Product Style Standard

Revision 1 status: Approved MVP project logic. Budget ranges and allocation behavior may be refined later using observed inventory, customer behavior and recommendation performance.